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High-temperature kinetics of propyne and allene: Decomposition vs. isomerization

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Abstract

The kinetics of the unimolecular dissociation of propyne and allene, $C_3H_4 + M \rightarrow C_3H_3 + H + M$, was investigated behind reflected shock waves at temperatures between 1400 and 2150 K and at pressures near 0.3, 1.3, 2.6 (propyne only), and 4.0 bar with argon as bath gas. Rate coefficients were obtained from the initial slope of the hydrogen-atom concentration–time profiles monitored with atomic resonance absorption spectroscopy at the Lyman α wavelength (121.6 nm). Within the experimental uncertainty (±30%), identical rate coefficients for propyne and allene decomposition were obtained, indicating a fast mutual isomerization. The dissociation reactions are shown to be in the low-pressure limit with a bimolecular rate coefficient $k^{\mathrm{bim}}(T) = (3.4 \pm 1.0) \times 10^{12} \exp[-(36,300 \pm 400)\ \mathrm{K}/T]\ \mathrm{bar}^{-1}\ \mathrm{s}^{-1}$. From a combination of our experimental results with kinetic data from the literature, we infer the following temperature and pressure dependence of the rate coefficient, which reproduces most of the experimental data at temperatures between 1200 and 2400 K and pressures between 0.1 and 5 bar better than within a factor of two: $k(T,P) = 2.58 \times 10^{41} (T/\mathrm{K})^{-7.81} \exp(-50,590\ \mathrm{K}/T)(P/\mathrm{bar})\ \mathrm{s}^{-1}$. This corresponds to a bimolecular rate coefficient in concentration units of $k^{\mathrm{bim}}(T) = 3.56 \times 10^{19} (T/\mathrm{K})^{-6.81} \exp(-50,590\ \mathrm{K}/T)\ \mathrm{cm}^3\ \mathrm{s}^{-1}$.
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1. Introduction

Propyne ($H_3C–C{\equiv}CH$, pC_3H_4) and allene ($H_2C{=}C{=}CH_2$, aC_3H_4) are important species in hydrocarbon combustion and pyrolysis, because they are precursors for the propargyl radical (C_3H_3), which is considered a critical intermediate in the formation of the first aromatic ring from aliphatic fuels [1–5]. The propargyl radical can be formed from propyne and allene in bimolecular hydrogen-abstraction steps (which requires a sufficiently high radical concentration) and by thermal unimolecular decomposition:

$$pC_3H_4 + M \rightarrow C_3H_3 + H + M \quad (1)$$

$$aC_3H_4 + M \rightarrow C_3H_3 + H + M \quad (2)$$

The kinetics of these reactions at higher temperatures is influenced, however, by a mutual thermal unimolecular isomerization

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$$pC_3H_4 + M \rightleftharpoons aC_3H_4 + M \quad (3)$$

and/or a bimolecular hydrogen atom-assisted isomerization [6–8]

$$pC_3H_4 + H \rightleftharpoons aC_3H_4 + H \quad (4)$$

Again, the latter reaction is only important, if hydrogen atoms are present in a sufficiently high concentration. Different aspects of the competition between the thermal unimolecular reactions (1)–(3) have attracted considerable attention over the years [1,6,7,9–21].

In a recent work of Miller and Klippenstein [21], the different experimental results are reviewed and analyzed in terms of a time-dependent, multiple-well master equation. The authors conclude from an eigenvector–eigenvalue analysis that at temperatures $T > 1800$ K (for pressures around 1 bar) a distinction between reactions (1) and (2) in kinetic experiments becomes increasingly difficult due to rapid interconversion of the reactants in reaction (3). This is in accord with conclusions from earlier works [1,6,7,9–11,13–18]. Convenient expressions for the rate coefficient of reaction (3) are given in [7], e.g., for a pressure of $P = 1.013$ bar: $k_3(T, P = 1.013\ \text{bar}) = 5.2 \times 10^{60}(T/\text{K})^{-13.93} \exp(-45,800\ \text{K}/T)\ \text{s}^{-1}$, and an analogous equation for the reverse reaction (−3) can be found in [21]: $k_{-3}(T, P = 1.013\ \text{bar}) = 7.8 \times 10^{39}(T/\text{K})^{-7.80} \exp(-39,478\ \text{K}/T)\ \text{s}^{-1}$. Both parameterizations reproduce most of the earlier experimental results with reasonable accuracy [7,21]. For example, for a temperature of $T = 2000$ K, one obtains $k_3 = 6.1 \times 10^4\ \text{s}^{-1}$ and $k_{-3} = 3.7 \times 10^5\ \text{s}^{-1}$, which results in a relaxation time of $(k_3 + k_{-3})^{-1} \sim 2\ \mu\text{s}$, being short compared to the timescale of typical shock-tube experiments in the order of 100–1000 μs.

In contrast to the large number of works on reaction (3), experimentally determined rate coefficients for reactions (1) and (2) are scarce [1,6,15,20]. In a most early work in 1987, Wu and Kern [1] from a shock-tube study of allene pyrolysis ($T = 1300$–2000 K, $P = 0.2$–0.5 bar, bath gas: Ne) inferred rate coefficients by modeling concentration–time profiles of different reaction products in terms of a complex mechanism consisting of 80 reactions. Identical Arrhenius expressions $k_1^{\text{bim}} = k_2^{\text{bim}} = 1.0 \times 10^{17} \exp(-35,200\ \text{K}/T)\ \text{cm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$ were employed, resulting in pseudo-first-order rate coefficients at $T = 2000$ K and $P = 1$ bar of $k_1 = k_2 = 1.4 \times 10^4\ \text{s}^{-1}$ considerably smaller than $(k_3 + k_{-3})$ given above.

Two years later, Hidaka et al. [15] studied propyne and allene pyrolysis in a shock tube ($T = 1200$–1570 K, $P = 1.7$ and 2.6 bar, bath gas: Ar) with time-resolved detection of the educts by IR emission and gas-chromatographic product analysis. Again a complex mechanism (34 reactions) was used for modeling, and the following temperature dependencies for the rate coefficients of reactions (1) and (2) were obtained from the fits: $k_1^{\text{bim}} = 4.7 \times 10^{18} \exp(-40,000\ \text{K}/T)\ \text{cm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$ and $k_2^{\text{bim}} = 2.0 \times 10^{18} \exp(-40,000\ \text{K}/T)\ \text{cm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$. This results in pseudo-first-order rate coefficients at $T = 2000$ K and $P = 1$ bar of $k_1 = 5.8 \times 10^4\ \text{s}^{-1}$ and $k_2 = 2.4 \times 10^4\ \text{s}^{-1}$, again smaller than $(k_3 + k_{-3})$ given above and with $k_1 > k_2$.

In a combined experimental and theoretical study, Kiefer et al. [6] determined rate coefficients for reactions (1) and (2) from shock-tube studies ($T = 1800$–2500 K, $P = 90$–900 mbar, bath gas: Kr or Ne) with laser-schlieren and time-of-flight mass spectrometric detection and performed a RRKM analysis. These authors obtained rate coefficients $k_1(T, P)$ and $k_2(T, P)$ very close to each other but with k_1 slightly larger than k_2 and with surprisingly little pressure dependence (i.e., close to first-order). The latter was ascribed to a comparatively high density of states of pC_3H_4/aC_3H_4 due to extensive isomerization-hindered internal rotation [6] (see also the discussions in [18]). Unfortunately, no compact parameterizations of k_1 and k_2 are given in [6]. For the sake of comparison, we derived the following approximate expressions from Figs. 8 and 9 of [6] (neglecting the weak pressure dependence): $k_1^{\text{bim}} = 7.3 \times 10^{16} \exp(-31,800\ \text{K}/T)\ \text{cm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$ and $k_2^{\text{bim}} = 1.1 \times 10^{17} \exp(-33,100\ \text{K}/T)\ \text{cm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$. From these Arrhenius expressions, we obtain for $T = 2000$ K and $P = 1$ bar: $k_1 = 5.5 \times 10^4\ \text{s}^{-1}$ and $k_2 = 4.3 \times 10^4\ \text{s}^{-1}$.

Finally, Scherer et al. [20] performed a shock-tube study ($T = 1300$–1700 K, $P = 1.7$–1.9 bar, bath gas: Ar), using atom resonance absorption spectroscopy to detect hydrogen atoms as a product of reactions (1) and (2). The following Arrhenius expressions were obtained: $k_1 = 1.34 \times 10^{12} \exp(-35,200\ \text{K}/T)\ \text{s}^{-1}$ and $k_2 = 2.3 \times 10^{12} \exp(-35,070\ \text{K}/T)\ \text{s}^{-1}$. From extrapolation to $T = 2000$ K and $P = 1$ bar, by assuming first-order pressure dependence, it follows: $k_1 = 1.7 \times 10^4\ \text{s}^{-1}$ and $k_2 = 3.1 \times 10^4\ \text{s}^{-1}$ with $k_2 > k_1$ in contrast to the results of [6,15].

As can be seen from our extrapolations to uniform conditions ($T = 2000$ K and $P = 1$ bar), all four experimental studies on reactions (1) and (2) appear to give grossly consistent kinetic data. However, it remains unclear whether reaction (1) or reaction (2) is faster, or if the isomerization reaction (3) is fast enough for both reactions to occur with virtually identical rates at least on practically relevant timescales. Since also no systematic study of the pressure dependence is available (apart from the study of Kiefer et al. for $T > 1800$ K and $P < 1$ bar [6]), and k_1 and k_2 were mostly obtained from complex modeling, we performed a shock-tube study over a comparatively broad pressure range and used direct detection of H atoms by atomic resonance absorption

spectroscopy (ARAS) for the determination of the rate coefficients. Furthermore, we tried to set the conditions for the experiments with propyne and allene as similar as possible to carefully quantify the differences. Our results will be discussed in the context of earlier experimental and theoretical work, and the rate coefficients will be parameterized for use in combustion models.

2. Experimental

The experiments were performed in a stainless steel shock tube behind reflected shock waves at temperatures between 1400 and 2150 K and pressures between 0.3 and 4.0 bar with Ar as the bath gas. The initial concentrations of propyne or allene were in the range 2–15 ppm. The detailed reaction conditions are listed in Tables S1 (allene) and S2 (propyne) of the Supplemental data. Since the experimental setup and methodology were described previously in detail ([8] and earlier references cited therein), we will only give a brief summary here.

The driver section and test section of the shock tube have lengths/diameters of 3.05 m/9.85 cm and 4.20 m/10.0 cm, respectively. The shock waves were initiated by pressure bursting of an aluminum foil separating the two sections, where hydrogen served as driver gas. The shock-wave velocity was measured with four pressure transducers, and the post-shock conditions were calculated from the initial temperature, pressure, and shock-wave velocity on the basis of one-dimensional conservation equations (see, e.g., [22]).

The hydrogen-atom concentration was monitored by ARAS at the Lyman α line (121.6 nm). The vacuum ultraviolet (VUV) radiation was generated in a microwave discharge lamp, transmitted via MgF_2 windows through the shock tube, and, after passing a VUV monochromator, detected with a solar-blind photomultiplier. The signal from the photomultiplier is sampled by a digital storage oscilloscope and further processed in a personal computer. The optical setup was calibrated with N_2O/H_2 mixtures as a well-characterized hydrogen-atom source [23].

The test gas and the calibration mixtures were manometrically prepared in two different stainless steel mixing vessels. The purity of the gases and chemicals used were as follows: propyne (Aldrich) > 98%, allene (Lancaster) > 97% (propyne and allene were degassed several times before using), H_2 for calibration (Messer Griesheim) > 99.999%, H_2 as driver gas (Air Liquide) > 99.8%, argon (Air Liquide) > 99.9999%, and N_2O (Messer Griesheim) 99%.

3. Results and discussion

A typical H atom concentration–time profile as obtained from our calibrated ARAS measurements is shown in Fig. 1 (here, exemplarily for propyne, experiments with allene gave analogous profiles). There is a fast increase caused by the decomposition of C_3H_4 and a leveling-off at later times caused by H-consuming bimolecular reactions. In modeling calculations, we could show that the most important H-consuming reactions under our conditions are $pC_3H_4 + H \rightarrow CH_3 + C_2H_2$ and $aC_3H_4 + H \rightarrow C_3H_3 + H_2$. Due to the low initial concentrations of C_3H_4, however, their influence during the first 100 μs can be neglected. We further note that the competing dissociation reactions $C_3H_4 + M \rightarrow C_3H_2 + H_2 + M$ were shown to be unimportant due to their higher threshold energies and much tighter transition states [21]. Accordingly, the rate coefficients k_1 and k_2 can be determined from the initial slopes of the profiles via the relation $k_i = [C_3H_4]_0^{-1} \times d[H]/dt$. The approach is illustrated in Fig. 1.

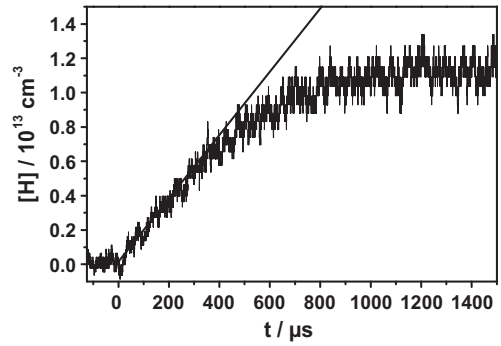

Fig. 1. Hydrogen-atom concentration–time profile detected by ARAS ($[pC_3H_4]_0 = 2.3 \times 10^{13}$ cm^{-3}, $[Ar] = 5.4 \times 10^{18}$ cm^{-3}, $T = 1620$ K, $P = 1.2$ bar), the solid line represents the initial slope ($k_1 = 7.9 \times 10^2$ s^{-1}).

The rate coefficients obtained in this way are displayed in Fig. 2, and the complete set of numer-

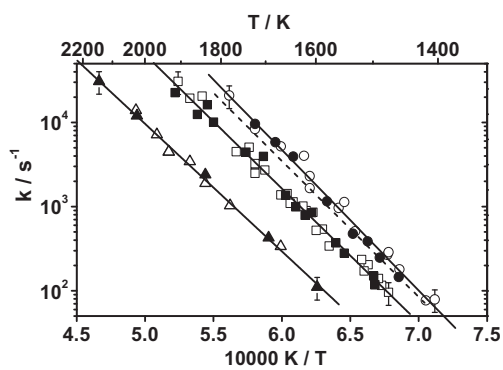

Fig. 2. Arrhenius plots of the experimental rate coefficients and linear least-squares fits, Eqs. (5)–(8) (full symbols: propyne, open symbols: allene, triangles: $P \sim 0.3$ bar, squares: $P \sim 1.3$ bar, circles: $P \sim 4.0$, dashed line: linear least-squares fit for additional propyne experiments at $P \sim 2.6$ bar).

ical values along with the detailed reaction conditions are collected in Tables S1 and S2 of the Supplementary data. The maximum error of k_1 and k_2 is estimated to be ±30%. We note that in Fig. 2 for the sake of clarity, experimental points for a pressure of $P \sim 2.6$ bar (only propyne experiments) are not shown (though contained in Table S1). The corresponding least-squares fit, however, is displayed as a dashed line.

We observe a pronounced temperature and pressure dependence of the rate coefficients, but within the uncertainty of our experiments, we do not see any significant difference between k_1 and k_2. That is, within the error margin of our experiments (±30%), propyne and allene decompose equally fast. The temperature and pressure dependence $k_1(T, P) = k_2(T, P) = k(T, P)$ can be represented by the following least-squares fits (cf. Fig. 2):

$$k(T, P \sim 0.3\,\text{bar}) = 3.6 \times 10^{11} \exp(-34{,}890\,\text{K}/T)\,\text{s}^{-1} \quad (5)$$

$$k(T, P \sim 1.3\,\text{bar}) = 5.9 \times 10^{12} \exp(-36{,}670\,\text{K}/T)\,\text{s}^{-1} \quad (6)$$

$$k(T, P \sim 2.6\,\text{bar}) = 1.7 \times 10^{13} \exp(-37{,}200\,\text{K}/T)\,\text{s}^{-1} \quad (7)$$

$$k(T, P \sim 4.0\,\text{bar}) = 4.9 \times 10^{13} \exp(-38{,}430\,\text{K}/T)\,\text{s}^{-1} \quad (8)$$

To check whether the rate coefficients are in the low-pressure limit, we calculated for each experimental point the bimolecular rate coefficient $k^{\text{bim}}(T, P) = k(T, P)/P$. The results are plotted in Fig. 3. It is evident that the bimolecular rate coefficients are virtually independent of pressure, which indicates that the reactions are indeed at or very close to the low-pressure limit. This is in contrast to the finding of Kiefer et al. [6,18], who obtained slightly pressure-dependent bimolecular rate coefficients in their laser-schlieren experiments, which seems to indicate that the rate coefficients are closer to the first-order regime, i.e., to the high-pressure limit. This somewhat unexpected behavior was already discussed by Kiefer et al. [6,18] and later in a theoretical study of Miller and Klippenstein [21]. Possible reasons can arise from anharmonicity effects in the density of states [6,18] but also from interferences of dissociation and isomerization at the short timescale characteristic for laser-schlieren experiments, which is in the order of a few microseconds [21]. Moreover, the laser-schlieren method is not sensitive to a special reaction, and additional information about the thermochemistry of all the occurring elementary steps is necessary for analysis (see the discussions in [6]). Obviously, the interference of dissociation and isomerization is not significant at the larger timescale of our experiments, and the direct observation of a primary reaction product allows data analysis without further assumptions. From an Arrhenius fit of our bimolecular rate coefficients (see Fig. 3), we obtained the following temperature dependence:

$$k^{\text{bim}}(T) = (3.4 \pm 1.0) \times 10^{12} \exp[-(36{,}300 \pm 400)\,\text{K}/T]\,\text{bar}^{-1}\,\text{s}^{-1} \quad (9)$$

To provide a parameterization, which is valid over a larger temperature and pressure range, we include in our further analysis the experimental results from [1,6,15,20]. On the basis of our observation that the rate coefficients under conventional shock-tube conditions are close to second order, we scaled the rate coefficients from [1,6,15,20] from their actual pressures to a pressure of 1.3 bar, where we performed most of our own experiments (see Fig. 2). This simple pressure scaling neglects the use of bath gases other than argon in [1] (neon) and [6] (krypton), but these effects are likely to be small compared to other uncertainties caused by different additional assumptions necessary for the complex data analyses in these works. Furthermore, we used averaged, pressure-independent expressions for the bimolecular rate coefficients from [6] (see Section 1). The result of our scaling is shown in Fig. 4.

With a certain exception in the case of the data from Wu and Kern [1], the match between the different expressions for $k(T)$ is satisfactory. In particular, the results in the upper temperature range of the allene experiments from Scherer et al. [20] and the propyne experiments from Hidaka et al. [20] coincide with the results in the lower temperature range of our experiments. The same applies to the data in the lower temperature range of the experiments of Kiefer et al. [6] and the data in the upper temperature range of our experiments. A

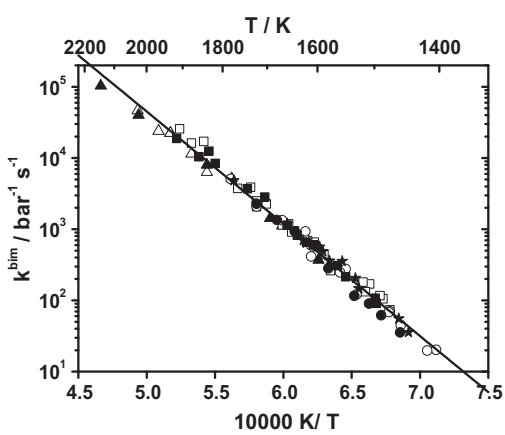

Fig. 3. Arrhenius plots of the bimolecular rate coefficients $k^{\text{bim}}(T, P) = k(T, P)/P$ and linear least-squares fit, Eq. (9) (symbols as in Fig. 2, stars: propyne data for $P = 2.6$ bar not included in Fig. 2).

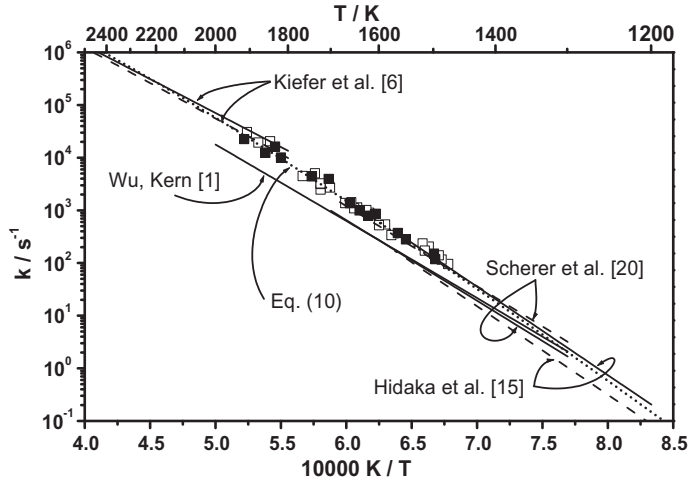

Fig. 4. Arrhenius plots of the rate coefficients k_1 and k_2 from [1,6,15,20] scaled to $P = 1.3$ bar (solid lines: k_1, dashed lines: k_2, Wu and Kern [1]: $k_1 = k_2$), symbols: our results for $P = 1.3$ bar, dotted line: Eq. (10) with $P = 1.3$ bar.

visual fit of these combined data and consideration of the first-order pressure dependence gives:

$$k(T,P) = 2.58 \times 10^{41} (T/\mathrm{K})^{-7.81} \times \exp(-50,590\ \mathrm{K}/T) \times (P/\mathrm{bar})\ \mathrm{s}^{-1} \quad (10)$$

which is shown as a dotted line in Fig. 4 for $P = 1.3$ bar. In Fig. 5, the adequacy of the pressure dependence of Eq. (10) is demonstrated in terms of our experimental results. The agreement is very good.

In view of these results, we recommend the use of Eq. (10) for the representation of the temperature and pressure dependence of the unimolecular dissociation reactions of propyne and allene, reactions (1) and (2), respectively, over a temperature range from 1200 K to 2400 K at pressures near 1 bar. The corresponding bimolecular rate coefficient in concentration units is

$$k^{\mathrm{bim}}(T) = 3.56 \times 10^{19} (T/\mathrm{K})^{-6.81} \times \exp(-50,590\ \mathrm{K}/T)\ \mathrm{cm}^3\ \mathrm{s}^{-1} \quad (11)$$

For most modeling purposes, this representation remains adequate (within a factor of 2) in the pressure range from ~0.1 to ~5 bar with increasing uncertainties at the limits of the temperature range due to falloff effects. Here additional experimental and/or theoretical work is necessary.

The use of identical rate coefficients for propyne and allene dissociation obviously relies on a comparatively fast mutual isomerization. In Fig. 6, the rate coefficient for the decomposition reaction is compared with theoretical predictions of the rate coefficients for the mutual isomerization [7,21]. One can realize that the isomerization reactions are faster than decomposition at least for temperatures below 2000 K. However, the timescale of isomerization also increases drastically with falling temperature, and the use of identical rate coefficients for decomposition is reasonable per se only for timescales longer than the temperature-dependent timescale of isomerization. That is, the typical timescale of the problem to be modeled has to be considered in using Eq. (10).

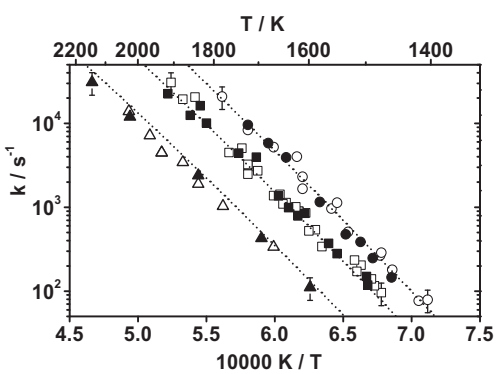

Fig. 5. Arrhenius plots of our experimental results and representation by Eq. (10) for $P = 0.3$ bar, 1.3 bar, and 4.0 bar.

4. Summary

The thermal unimolecular dissociation reactions of propyne and allene at temperatures between 1400 and 2150 K and pressures between 0.3 and 4 bar were studied in a shock tube by monitoring hydrogen atoms with ARAS. Within

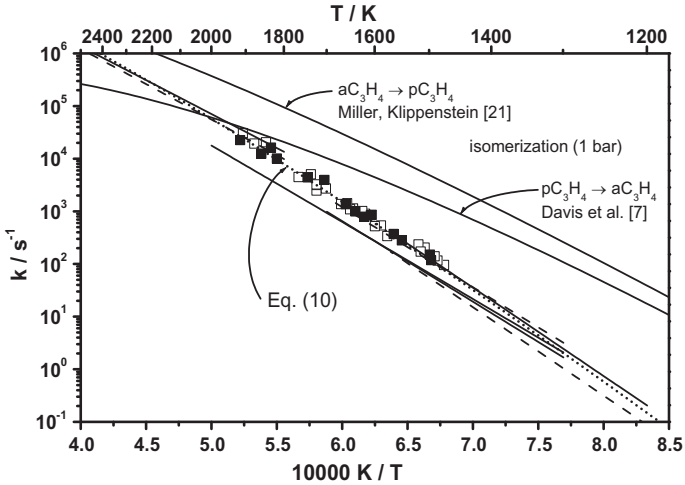

Fig. 6. Same as Fig. 4 but with rate coefficients for isomerization, $pC_3H_4 \rightleftharpoons aC_3H_4$, for $P = 1$ bar from [7,21].

the experimental uncertainty, identical rate coefficients were obtained, indicating that mutual isomerization is faster than unimolecular dissociation. The rate coefficients of the dissociation reactions are in the low-pressure limit, and a parameterization of the temperature and pressure dependence for use in combustion modeling is given.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.072.

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